Cyber Project – Task 1: Network Scanner using Python & Nmap

Introduction

This project focuses on developing a Python-based Network Scanner using the Nmap library in Kali Linux. Network scanning is an essential cybersecurity technique used to identify active hosts, discover open ports, and detect running services on a target system. The scanner automates the scanning process, making it easier to collect network information and save the results in a structured JSON format. This project provides practical exposure to Python scripting, network reconnaissance, and cybersecurity fundamentals.

Objective

The main objective of this project is to build a simple and efficient Network Scanner capable of scanning a target IP address using Python and Nmap. The scanner identifies available services and stores the scan results in a JSON file for future analysis. The project also aims to improve understanding of cybersecurity concepts, automation, and network reconnaissance techniques.

Libraries Used

- Python 3
- Nmap (python-nmap)
- JSON
- Oracle VirtualBox
- Kali Linux

Methodology

The project was implemented by first configuring the Kali Linux virtual machine and installing the required Python libraries. A Python script was developed using the python-nmap library to scan a target IP address. The scanner collects information about the available host and network services. The collected data is then organized into a structured dictionary and exported as a JSON file using Python's built-in JSON module. Finally, the project files, screenshots, and documentation were uploaded to GitHub for version control and project presentation.

Result

The Network Scanner successfully scanned the specified target IP address and generated the scan results in JSON format. The project demonstrated successful integration of Python with Nmap for automated network scanning. The generated JSON file can be used for reporting, analysis, and future cybersecurity tasks.

Conclusion

The Python-based Network Scanner was successfully developed and tested using Kali Linux and Nmap. This project enhanced my practical knowledge of network reconnaissance, Python programming, JSON data handling, and GitHub documentation. It also provided hands-on experience with cybersecurity tools and strengthened my understanding of automated network scanning techniques. The project serves as a strong foundation for learning advanced cybersecurity concepts and penetration testing.